

Peter Ziegler

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EDUCATION

University of Georgia, College of Engineering – Athens, GA Expected Dec 2027

B.S. Mechanical Engineering & Certificate in Artificial Intelligence GPA: 3.51

Skills:

Engineering Software: Onshape, AutoCAD LT, Autodesk Inventor Pro, MQTT, VT-Scada, Microsoft Office

Programming & Fabrication: MATLAB, Python, Arduino, Node-Red, Linux, FDM 3D Printing, MIG & Stick Welding

Relevant Coursework: Biomechanics, Data Science I, Circuits, Thermodynamics, Strength of Materials, Fluid Mechanics, Numerical Methods

PROFESSIONAL EXPERIENCE

Undergraduate Research Assistant, UGA Physical AI Lab – Athens, GA Jan 2026 – Present

- Ran a data-driven study testing whether collecting targeted subtask data improves efficient data collection compared to recording full task demonstrations, fine tuning Vision Language Action models across 4 dataset configurations.
- Designed a controlled data-collection environment isolating lighting and background variables, improving model success rate by 45%.
- Designed, iterated on, and 3D-printed a custom end effector that expanded grippable surface area 150%, improving grasp consistency in manipulation trials.

Teaching Assistant, UGA ECE Department – Athens, GA May 2026 – Present

- Scaled a non-majors circuits course from ~40 to ~175 students by co-developing a library of 13 hands-on demo videos that connect circuit theory to real-world electronics.
- Built demonstrations spanning appliance and motor tear downs (identifying transformers, capacitors, and armature/stator components), RL and RC low-pass filters with phase-difference measurement, and a digital logic circuit simulating a computer's memory.
- Designed a custom audio-amplification PCB in KiCad and fabricated the board. Produced a demo video on the full schematic-to-assembly workflow, shown in class and added to the course demo library.

Automation Engineering Intern, Southern Flow – Alpharetta, GA May 2025 – Aug 2025

- Designed the mechanical and pneumatic systems for a glycerin fill station, then built and verified a prototype against performance requirements, projected to reduce subcontracted labor and cost by 40%.
- Built and deployed a pilot Remote Telemetry Unit transmitting temperature and humidity data over MQTT, then documented the full fabrication, integration, and configuration so the design could be replicated across sites.
- Produced AutoCAD wiring diagrams mapping Allen-Bradley PLC I/O cards for four customer installations, translating field wiring into clear, standardized schematics.

PROJECT EXPERIENCE

Biomechanics, The University of Georgia – Athens, GA Jan 2026 – May 2026

- Co-developed a 6-link 2D static-equilibrium model in MATLAB to compute lower-back muscle force across 20 lifting postures, finding an upright squat cut peak force ~79% versus a forward stoop.
- Built a 2D inverse-dynamics solver in MATLAB that computes joint forces and torques from motion-capture and force-plate data, showing the ankle dominates push-off at 55% higher torque than the hip and 110% higher than the knee.

Inverse Kinematics Solver, Self Study Dec 2025 – Jan 2026

- Built a Python inverse-kinematics solver for a 6-axis UR5 arm that converges to a target pose in ~9 iterations, using a Newton-Raphson method on the Jacobian Matrix to hit a 1e-4 tolerance.
- Validated the solver across 50 random starting configurations with 100% convergence, verifying joint motion against a Coppeliasim robotics simulation.

Dart Launcher, The University of Georgia – Athens, GA Aug 2024 – Dec 2024

- Led fabrication and iterative testing on a 5-engineer team building a 5ft³ elastic-powered dart launcher, tuning trigger geometry, launch angle, and projectile stability to push launch distance 30% past baseline.

AWARDS & CERTIFICATIONS

Summer Research Award, The University of Georgia – Athens, GA May 2026

Lean Six Sigma Yellow Belt, AIGPE June 2026

Modern Robotics, Course 2: Robot Kinematics, Coursera Jan 2026